

The Special Theory of Computational Relativity: A Formalization and Audit

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Abstract

We recast the axiomatic monograph *The Special Theory of Computational Relativity* (v1.0.0) as a sequence of precisely stated definitions, propositions and proofs, and we record for each claim whether it is (i) a correct, standard result, (ii) true only by definition, (iii) true only conditionally on an added assumption, or (iv) not established. Euler’s identity and the counts 2^N and $\binom{N}{2}$ are proved. The relation $E = mc^2$ holds only as a definition (and only in the zero-momentum frame). The claimed constructive proof of $P = NP$ does *not* follow; we state the strongest correct replacement, a model-relative collapse that is trivial once its oracle axiom is granted. Consequently this paper does not prove $P = NP$.

1 Scope and summary of claim status

The source monograph asserts nine results. Table 1 gives the verdict established in the sections indicated.

Claim	Status	Where
2^N orthants, $\binom{N}{2}$ rotation planes	Proved (standard)	Prop. 1
Emergent metric g_{ij}	Defined; Riemannian only	Prop. 2
Euler’s identity	Proved (standard)	Prop. 3
$E = mc^2$ from field density	Tautological; frame-dependent	Props. 4, 5
Dimension–property scaling, constants	Definitions; no constants derived	§2
Orbital gear / firewall / time reversal	Not established; formula inconsistent	§5
$P = NP$ in $\mathcal{M}_N(A_0)$	Not established; conditional trivial version only	Props. 6, 7
P_{HSAM} qualia postulate	Ill-defined; unfalsifiable as stated	§7
Brass cube geometry	Dimensionally consistent; no link to theory	§8

Table 1: Status of claims.

2 Axiom A_0 and dimensional scaling

Axiom 1 (A_0 , as stated). $0 = 0$ and $\lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in \mathbb{R}^N$.

Audit. For every $x \neq 0$, $0/x = 0$, hence $\lim_{x \rightarrow 0} 0/x = 0$; the limit is not an unconstrained λ . The symbol $0/0$ is undefined in $\mathbb{R}$ (it is not “indeterminate” as a number; *limits of forms* f/g with $f, g \rightarrow 0$ may take any value, but the constant function 0 yields 0). A seed λ can only enter as an independent postulate, so we restate:

Axiom 2 (A'_0). There is a distinguished point O and a smooth map $\lambda : \mathcal{M} \rightarrow \mathbb{R}^N$ with $\lambda(O) = 0$.

Definition 1 (Property count). $P_N := 3 + N$ for $N \in \mathbb{Z}_{\geq 0}$, and $P_{-N} := 1/(3 + N)$ for $N \in \{1, 3, 5\}$ (giving $\frac{1}{4}, \frac{1}{6}, \frac{1}{8}$ as in the source).

Audit. This is a labelling; “properties” and “negative dimensions” are undefined, nothing is derived from them, and the “crystallization” $P_N \otimes P_{-N} \mapsto \{\delta_c, \phi_c, \mathcal{O}_8\}$ is never given an operation or a numerical value. No physical constant is obtained. Note also that $P_N \cdot P_{-N} = 1$ identically, so the collision of a property with its inverse is the multiplicative identity, not a constant. The symbols δ, ϕ are also reused for orbital tiers (§5).

3 Combinatorial and geometric content

Proposition 1. Let N orthogonal coordinate hyperplanes through O cut $\mathbb{R}^N$. Then (a) there are 2^N open orthants; (b) $\dim \mathfrak{so}(N) = \binom{N}{2}$, the number of coordinate 2-planes.

Proof. (a) An orthant is fixed by a sign choice $\pm$ for each of the N coordinates: 2^N choices. (b) The coordinate planes are the 2-subsets $\{i, j\} \subset \{1, \dots, N\}$; there are $\binom{N}{2} = N(N-1)/2$. The generators $L_{ij} = e_i e_j^T - e_j e_i^T$, $i < j$, form a basis of the skew-symmetric matrices. $\square$

Proposition 2. Let $J = D\lambda$ and $g_{ij} = \sum_{k=1}^N \partial_i \lambda_k \partial_j \lambda_k = (J^T J)_{ij}$. Then g is positive semi-definite, and positive definite exactly where J has rank $\dim \mathcal{M}$.

Proof. $v^T g v = \|Jv\|^2 \geq 0$, with equality iff $Jv = 0$. $\square$

Audit. (i) The additional term $\lim_{x \rightarrow 0} (0/x) g_{ij}^{(0)}$ equals 0 and drops out. (ii) g is positive definite, i.e. Riemannian. Special relativity requires Lorentzian signature $(-, +, +, +)$; nothing here produces a light cone, so the construction does not yield a *relativistic* geometry.

Proposition 3 (Euler’s identity). $e^{i\pi} + 1 = 0$.

Proof. Define $e^z = \sum_{k \geq 0} z^k/k!$ (absolutely convergent on $\mathbb{C}$). For $z = i\theta$, $\theta \in \mathbb{R}$, split even and odd k : $e^{i\theta} = \sum_m (-1)^m \theta^{2m}/(2m)! + i \sum_m (-1)^m \theta^{2m+1}/(2m+1)! = \cos \theta + i \sin \theta$. Define π as twice the smallest positive zero of $\cos$; then $\cos \pi = -1$, $\sin \pi = 0$ and $e^{i\pi} = -1$. $\square$

Audit. The result is a theorem of analysis, independent of A_0 and of any manifold; the source’s “derivation” (evaluating $\cos \pi + i \sin \pi$) presupposes $e^{i\theta} = \cos \theta + i \sin \theta$. The “anti-phase equilibrium across the mirror axis” is an interpretation, not a derived physical statement.

4 Mass–energy equivalence

Let $u = \frac{1}{2}(\epsilon_0\|\mathbf{E}\|^2 + \mu_0^{-1}\|\mathbf{B}\|^2)$, $\rho_m := u/c^2$, $m := \iiint_{\Omega} \rho_m d^3\mathbf{r}$.

Proposition 4 (Tautology). $\iiint_{\Omega} u d^3\mathbf{r} = mc^2$.

Proof. $\iiint u = \iiint c^2 \rho_m = c^2 m$ by linearity of the integral. $\square$

Audit. This follows from the definitions of ρ_m and m ; it is not a derivation of Einstein’s relation, since m was *defined* as E/c^2 . Whether this m is the inertial mass is a separate, physical question:

Proposition 5. *For a field with total energy E and momentum $\mathbf{P} = \epsilon_0 \iiint \mathbf{E} \times \mathbf{B} d^3\mathbf{r}$, the invariant mass is $m_{\text{inv}} = \sqrt{E^2 - \|\mathbf{P}\|^2 c^2}/c^2$. Thus $m = E/c^2 = m_{\text{inv}}$ iff $\mathbf{P} = 0$.*

Proof. $(E/c, \mathbf{P})$ is a four-vector for a closed system; its Minkowski norm is Lorentz invariant and equals $m_{\text{inv}} c$. $\square$

Audit. A single vacuum plane-wave pulse has $E > 0$ and $\|\mathbf{P}\|c = E$, so $m_{\text{inv}} = 0$ although the source’s $m > 0$; the source’s m is frame-dependent. Moreover, (i) for bound charge distributions the electromagnetic energy–momentum alone is not a four-vector (the classical “4/3 problem”; non-electromagnetic stresses are required), and (ii) vacuum Maxwell theory is linear and its finite-energy solutions with compact data disperse, so the hypothesis that matter is *purely* continuous EM field density is not supported by anything in the monograph. It also conflicts with discrete, quantized matter at the level of experiment (e.g. atomic spectra, Compton scattering).

5 Orbital gears, firewalls, time reversal

Correct content. A diatomic bond orbital with axial quantum number $|m_l|$ has $2|m_l|$ lobes and $|m_l|$ nodal planes containing the bond axis: δ ($|m_l| = 2$): 4 lobes, 2 planes; ϕ (3): 6, 3; $|m_l| = 4$ (conventionally γ): 8, 4. The source’s counts are consistent with this, but its “ $L = 2n = 2|m_l|$ ” conflates the lobe number with an angular-momentum symbol.

Not established. The source’s

$$\Psi_{\text{mesh}} = \sum_n (-L_n) \prod_{j=1}^{N_{\text{node}}} (1 - \Theta_{\text{firewall}}(x_j)) \exp\left(-\frac{\Delta E_{\text{Landauer}}}{k_B T}\right)$$

has these defects:

1. Θ_{firewall} is undefined. If it equals 1 at any node (i.e. a firewall is present, as the text asserts), the product vanishes and $\Psi_{\text{mesh}} \equiv 0$: the expression annihilates itself.
2. $|\psi| = 0$ on a node does not make it impermeable. Probability current $\mathbf{j} \propto \text{Im}(\psi^* \nabla \psi)$ is generally nonzero across the nodes of one component of a superposition, and nodes of time-dependent states move.
3. Landauer’s bound $\Delta E \geq k_B T \ln 2$ is a lower bound on heat dissipated when erasing a bit in a thermal reservoir; geometric nodal surfaces neither absorb it nor isolate “timelines”. Unitary quantum dynamics is already reversible and creates no cross-time “bleeding”.
4. Quantum time reversal is an antiunitary operator; “inverting torque vectors” is a classical statement, and the exponential factor $e^{-\Delta E/k_B T}$ has no derivation as a transmission amplitude.

6 The $P = NP$ claim

Source theorem. In $\mathcal{M}_N(A_0)$, evaluation of path candidates over 2^N hyper-octants reduces to $O(N^2)$ algebraic minimization, hence $P = NP$.

Why the argument fails. Steps refer to the source’s proof.

1. *Step 1* asserts $\mathcal{S} \rightarrow \min_k E(x_k)$ without specifying the index set of k . If k runs over 2^N states the minimum is not computed by the argument; if the landscape is arbitrary, ground-state problems (e.g. non-planar Ising/spin glass) are NP-hard (Barahona, 1982). Physical relaxation reaches local minima, not guaranteed global ones.
2. *Steps 2–3* count planes, $\binom{N}{2}$, not candidates; $\binom{N}{2} \ll 2^N$ for large N . Evaluating 2^N candidates via $\binom{N}{2}$ evaluations needs a map from planes to candidates, which is not given. Also, “exactly $\frac{1}{2}(N^2 - N)$ operations” counts planes; the cost per plane is unspecified.
3. *Step 4* sets $\mathcal{T}_{\text{solve}} = 0$, contradicting the $O(N^2)$ claimed in Step 5; zero time also has no meaning in a computational model, and it rests on the unestablished firewall of §5.
4. *No reduction.* A proof of $P = NP$ requires an explicit polynomial-time algorithm for an NP-complete problem (e.g. 3-SAT) on a Turing-equivalent model. None is encoded here.

Proposition 6 (Query lower bound). *Given black-box access to $f : \{0, 1\}^N \rightarrow \{0, 1\}$ with a single unknown $f(w^*) = 1$, any classical algorithm needs $\Omega(2^N)$ queries to find w^* with constant success probability, and any quantum algorithm needs $\Omega(2^{N/2})$ (Bennett–Bernstein–Brassard–Vazirani, 1997).*

Hence $O(N^2)$ evaluations cannot search 2^N candidates unless the structure of the instance is exploited, and no such structure is exhibited.

Proposition 7 (Model-relative collapse). *Let MIN be an oracle that, given a polynomial-time verifier V and input x , returns $\min_{w \in \{0, 1\}^{p(|x|)}} E_x(w)$ with $E_x(w) = 0$ if $V(x, w) = 1$ and 1 otherwise, at unit cost. Then $NP \subseteq P^{\text{MIN}}$.*

Proof. For $L \in NP$ with verifier V , $x \in L \iff \min_w E_x(w) = 0$: one oracle query. $\square$

Interpretation. This is the strongest correct reading of “ $P = NP$ in $\mathcal{M}_N(A_0)$ ”: the collapse is exactly the assumption that the model supplies a unit-cost global-minimization primitive. It is true for the trivial reason that MIN decides an NP-complete set. It says nothing about P vs. NP in the Turing sense, since relativized collapses are known ($P^A = NP^A$ for some oracles A ; Baker–Gill–Solovay, 1975), as are collapses for machine models with unrealistic unit-cost arithmetic (Hartmanis–Simon, 1974). Whether any physical system implements MIN is an empirical question the monograph does not address; known physical heuristics (annealing, adiabatic evolution) are not known to do so.

7 Postulate P_{HSAM}

The postulate $\mathcal{C}(\vec{t}) = \|d^2\vec{t}/d\tau^2\| = \mathbf{Q}$ is ill-posed: time is a coordinate (a scalar function), not a vector field, and for an observer worldline $t(\tau)$ one has $d^2t/d\tau^2 = d\gamma/d\tau$. This is frame-dependent and vanishes for every inertial observer, so, as written, inertial observers have no qualia. It maps no measurable neural or behavioural quantity onto $\mathbf{Q}$, so it is untestable. The claim that HSAM is “lossless” is contradicted by data: HSAM subjects show normal-range susceptibility to false memories (Patihis et al., 2013).

8 The brass sculpture

Dimensional checks: the three shells 300, 200, 100 mm differ by 50 mm per wall coordinate, so each corner pillar spans $50\sqrt{3} \approx 86.6$ mm between adjacent shells; apertures $\varnothing 220 < 300$, $\varnothing 140 < 200$, $\varnothing 70 < 100$ leave positive rims; a 300 mm CuZn39Pb3 billet ($\rho \approx 8.45 \text{ g cm}^{-3}$) has mass ≈ 228 kg. The corner count $8 = 2^3$ is the number of cube vertices, not evidence about $\mathcal{M}_N(A_0)$. Machinability of the enclosed cavities through the apertures (tool reach) is not demonstrated and should be checked with a CAM model. The sculpture is an artistic realization; it is not a test of any claim above.

9 Conclusion

Proved: Prop. 1 (counts), Prop. 3 (Euler), Prop. 2 (Riemannian metric from a seed map), Prop. 7 (oracle-relative collapse). Definitional only: the $E = mc^2$ identity of §4. Not established: $P = NP$ in any sense relevant to the open problem, field-only matter, constant crystallization, nodal firewalls, and P_{HSAM} . To strengthen the programme, the minimal missing items are: a Lorentzian metric; a defined Θ_{firewall} ; an explicit encoding of 3-SAT into $\mathcal{M}_N(A_0)$ with a polynomial-time algorithm; and a numerically falsifiable prediction. The “Observation Shearing Operator $\hat{\mathcal{S}}$ ” listed in the source’s traceability matrix is never defined in the text and is omitted here.

References

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